

ENGINEERING PROPERTIES OF ROCK MASS

Laboratory sampling and testing

Exact knowledge of the mechanical properties of intact rock requires laboratory testing of representative samples of intact rock. The following tests were considered necessary:

- **Uniaxial compressive strength, UCS**
- **Modulus Elasticity, E**
- **Point load Index, $I_{s(50)}$**
- **Brazilian tensile strength, BTS**
- **Shear strength**
- **Seismic Velocity(Surface & Body waves)**

ENGINEERING PROPERTIES OF ROCK MASS

- **Petrographic analysis/thin section**
- **Drilling rate index/Cutter life index (DRI/CLI)**
- **Abrasivity (Cerchar scratch test)**
- **Slake durability Index**
- **Density**
- **Porosity**

LABORATORY TESTS

Laboratory testing methods for soil/rock material are generally well established and testing techniques have been recommended by the International Society for Rock Mechanics (ISRM) or Bureau of Indian standards (BIS).

Many of the large field tests are very expensive. As a consequence, the value of results found from such tests should be carefully weighted against the costs.

ENGINEERING PROPERTIES OF ROCK MASS

The strength of the rock mass is controlled by the complex interaction of a number of factors including:

- Intact rock strength
- Geological structure
- Groundwater
- Alteration of minerals on exposure to air and/or water with time.

ENGINEERING PROPERTIES OF ROCK MASS

As a result of the complex interaction of the above factors, that can occur when rock is subject to load, it has also been found that the strength of rock, in general, is dependent on the volume of rock being loaded and the direction in which the load is applied.

ENGINEERING PROPERTIES OF ROCK MASS

This volume and directional dependence of rock strength is not generally found in other engineering materials.

Rock mass strength is the least well defined aspect of geotechnical engineering. There is a need to have a much better understanding of rock mass strength.

ENGINEERING PROPERTIES OF ROCK MASS

Back analysis, typically of instrumented sections (e.g. In situ test) of an excavation site and/or failures, can be a very useful approach to estimating the rock mass strength. The result of in situ test relies on instrumentation to determine, directly or by calculation, changes in displacements, strains, pressures and stresses during excavation.

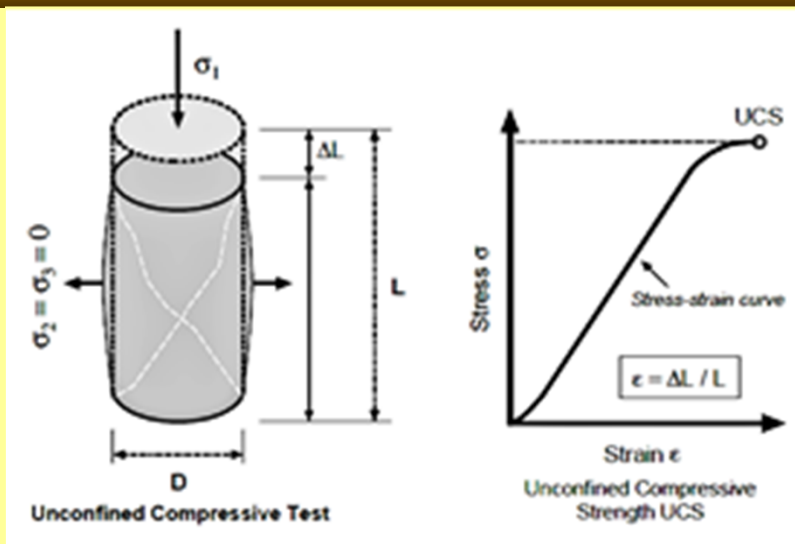
ENGINEERING PROPERTIES OF ROCK MASS

This approach generally requires a good knowledge of the geometry of the situation, stress field, likely mode of failure, influence of geological structure, use of appropriate numerical model(s), etc for success.

ENGINEERING PROPERTIES OF ROCKS

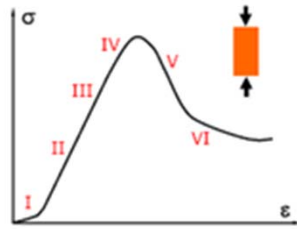
- The strength of a rock may be established either experimentally by means of laboratory testing of intact rock specimens, and/or by rock-testing in situ.
- Important mechanical (strength) properties of rocks are:
 - Compressive strength
 - Uniaxial or unconfined compressive strength
 - Triaxial compressive strength
 - Tensile strength
 - Direct test
 - Indirect method (Brazilian test)
 - Direct shear strength

COMPRESSIVE STRENGTH



Unconfined compressive test - core specimen with failure under unconfined compression

COMPRESSIVE STRENGTH



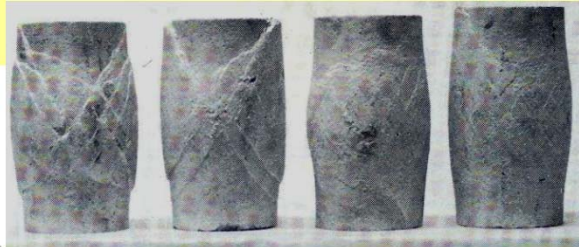
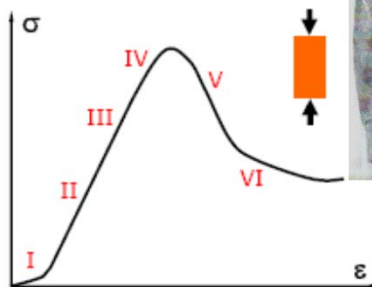
Around peak stress (IV-V)



Post peak stress (VI)

Typical uniaxial compression stress-strain curve of rock material.

Uniaxial Compressive strength



COMPRESSIVE STRENGTH

The ultimate strength of rock material is characterized by the stress value at failure given by:

$$\sigma_c = \frac{P}{A} = \frac{P}{\pi/4 D^2}$$

Where P = failure load

A = initial cross-sectional area at mid-height of the specimen

D = diameter at mid-height of the specimen

σ_c = uniaxial compressive strength of rock

COMPRESSIVE STRENGTH

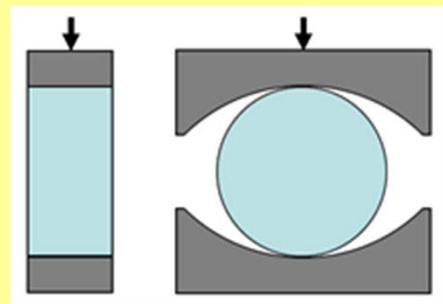
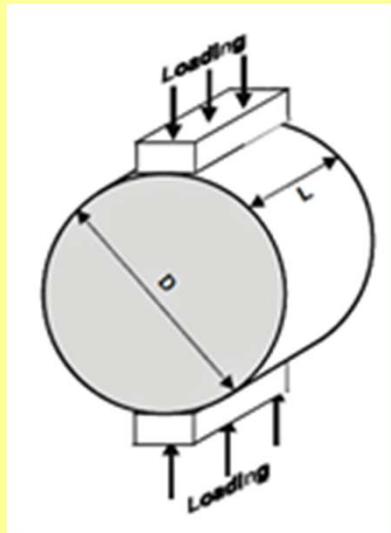
Classification of Rock Materials for Strength (Bieniawski, Z. T)

Description	Uniaxial Compressive Strength, MPa
Very High Strength	> 200
High Strength	100 - 200
Medium Strength	50 - 100
Low Strength	25 - 50
Very Low Strength	< 25

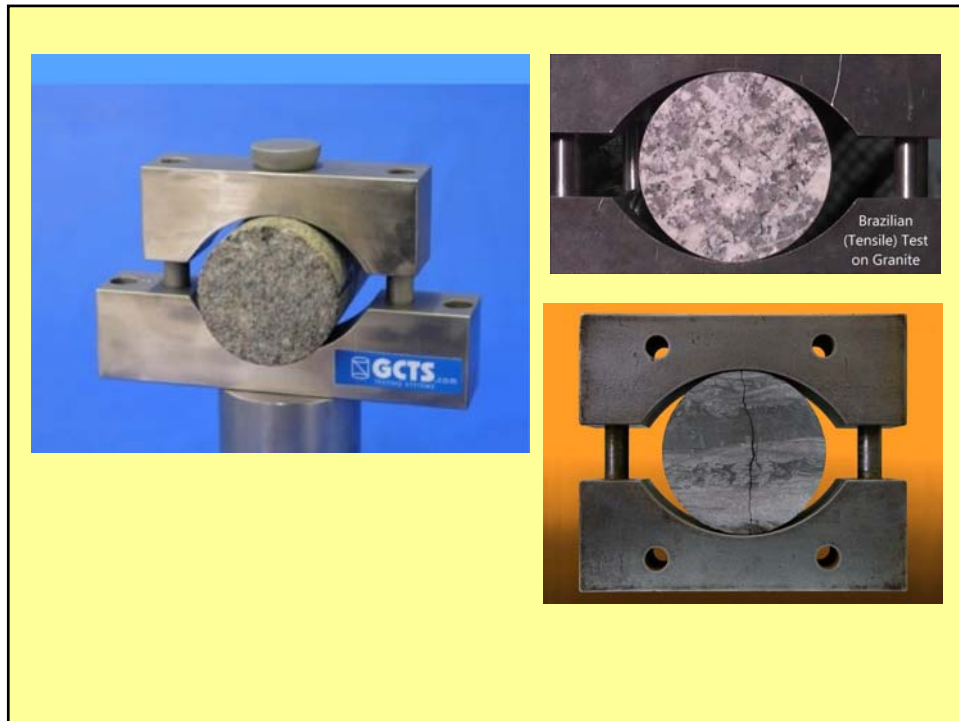
TENSILE STRENGTH

- Tensile strength of rock materials can be obtained from several types of tensile tests:
 - direct tensile test,
 - Brazilian test and
 - flexure test.
- Direct test is not commonly performed due to the difficulty in sample preparation.

TENSILE STRENGTH



Brazilian testing arrangement for determining indirect tensile strength. Core specimen



TENSILE STRENGTH

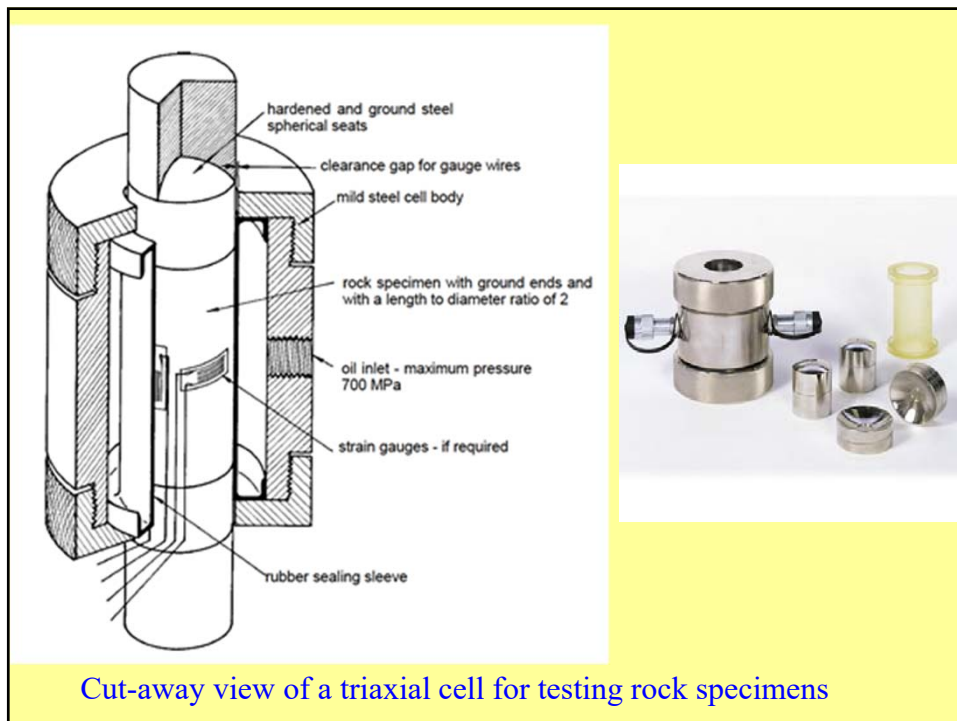
In the center of the disc the tensile stresses are given by

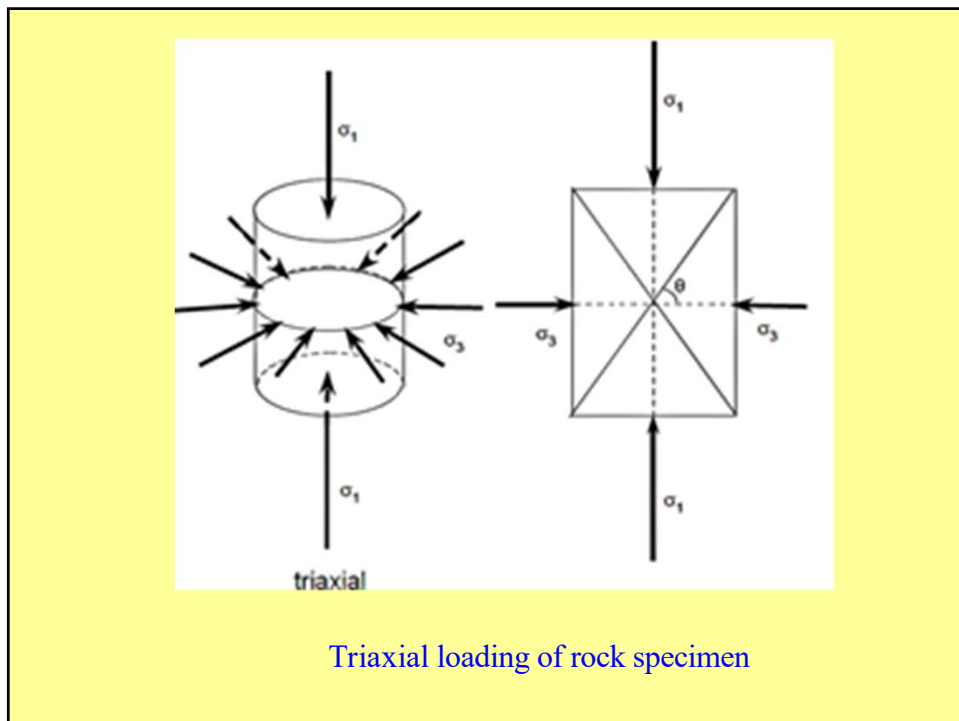
$$\sigma_t = \frac{2P}{\pi Dt}$$

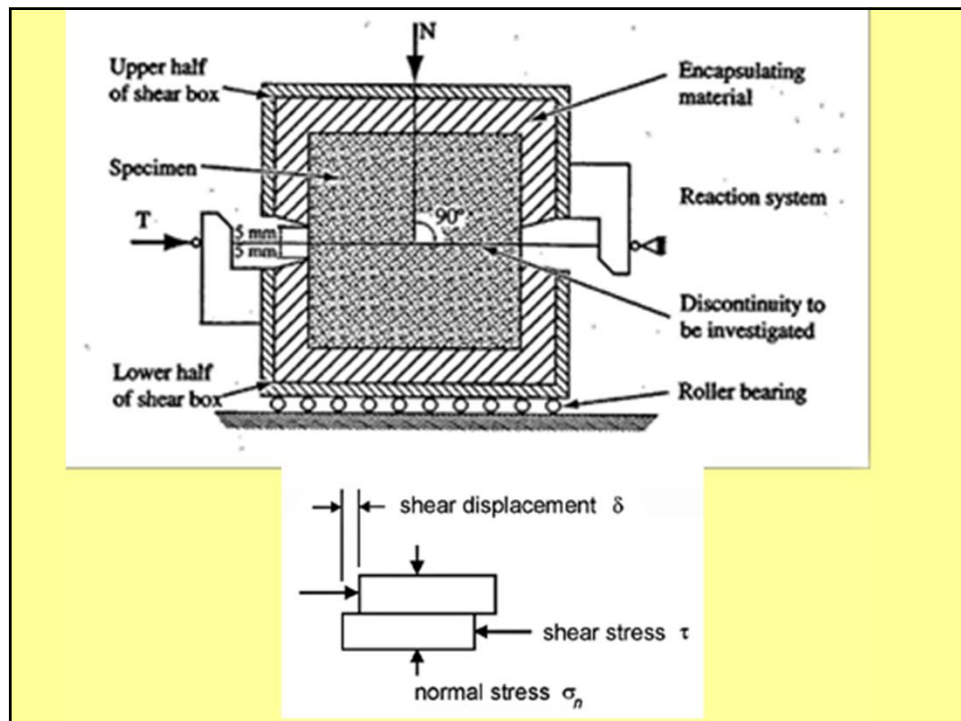
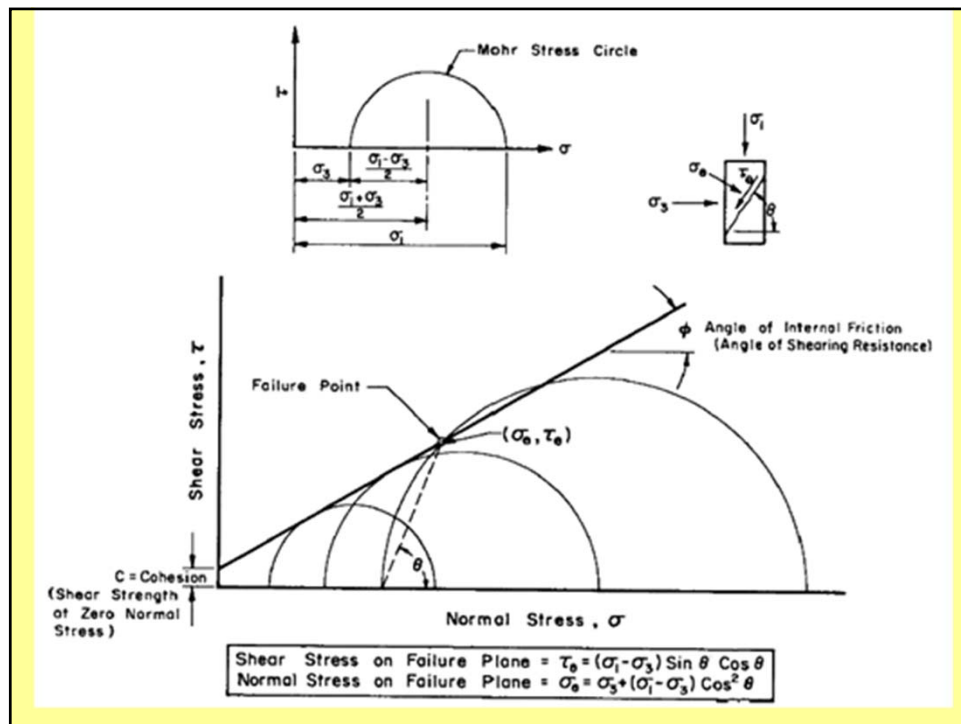
Where, P = applied load at failure, MN
 D = diameter of the disc (m)
 t = thickness of the disc (m)
 σ_t = tensile strength of rock (MPa)

TRIAXIAL STRENGTH

- Triaxial compressive strength, is calculated as the axial failure load divided by the initial cross sectional area of the specimen.
- With a series of triaxial tests conducted at different confining pressures, peak stresses (σ_1) are obtained at various lateral stresses (σ_3).
- By plotting Mohr circles, the shear envelope is defined which gives the cohesion and internal friction angle.







SHEAR STRENGTH (DIRECT SHEAR TEST)

The stress values were calculated by dividing the loads by the nominal areas (initial cross-sectional areas) of the interfaces.

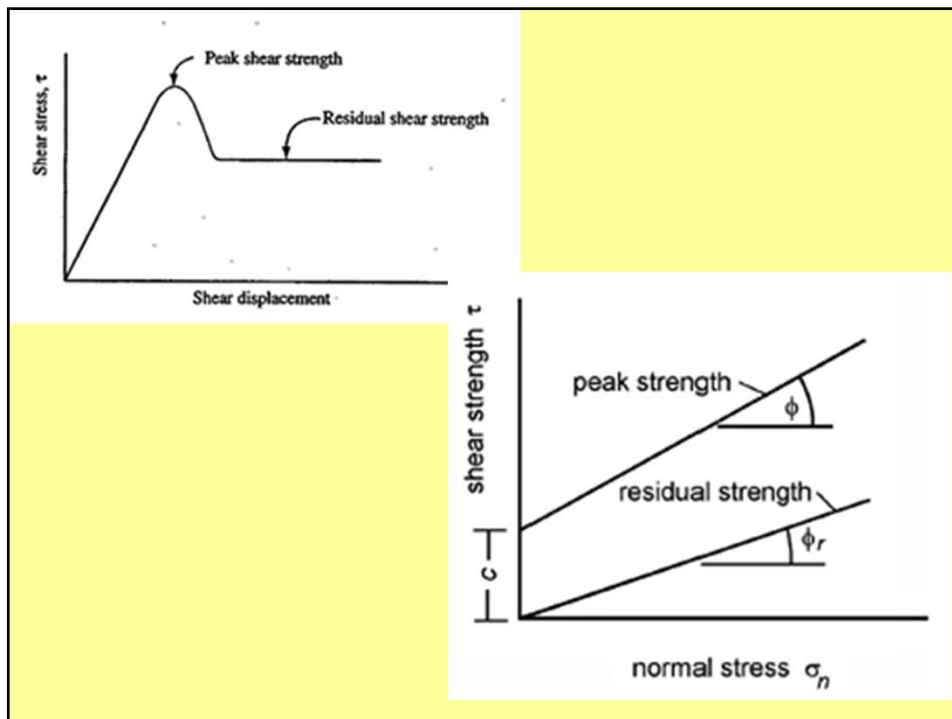
Normal Stress, $\sigma_n = P_n/A$, MPa

Shear Stress, $\tau = P_s/A$, MPa

where: P_n = normal load, kN

P_s = shear load, kN

A = nominal or apparent cross sectional area, mm^2

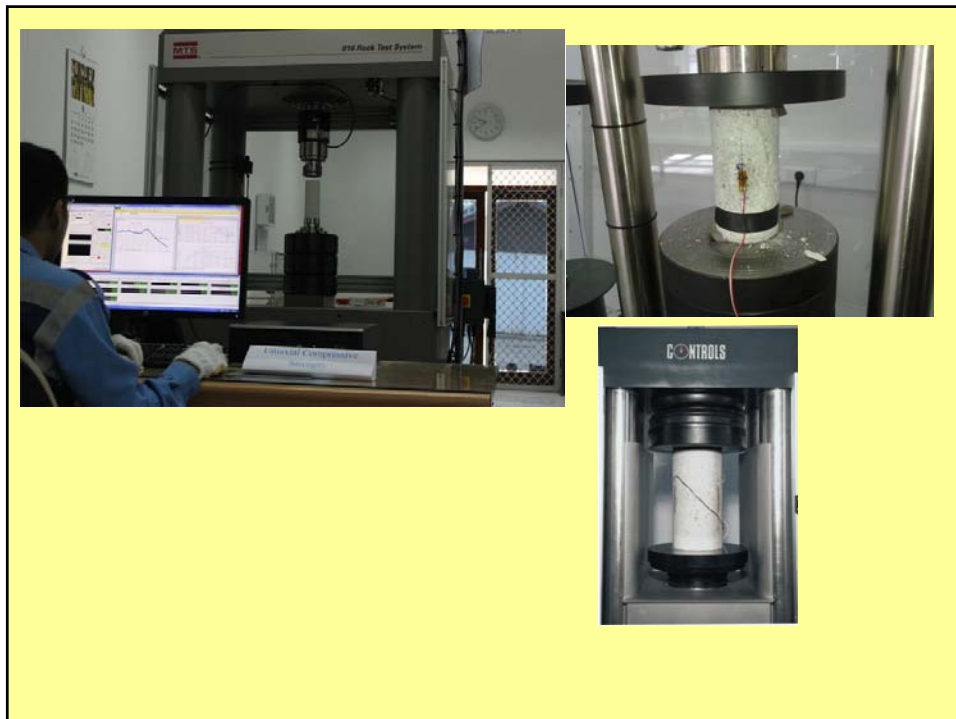


SHEAR STRENGTH (DIRECT SHEAR TEST)

The relationship between the peak shear strength τ_p and the normal stress σ_n can be represented by the Mohr-Coulomb equation:

$$\tau_p = c + \sigma_n \tan \phi$$

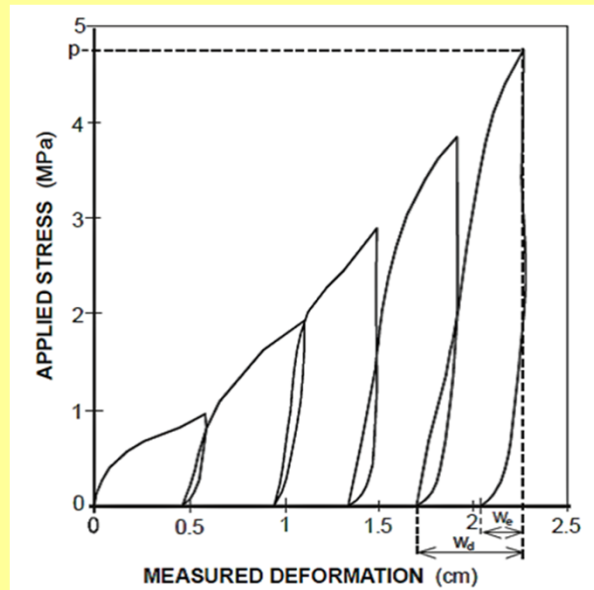
Where c is the cohesive strength of the cemented surface and ϕ is the angle of friction.



FIELD TESTS

Various field investigations and tests, such as:

- ❖ shear strength measurements of jointed rockmass
- ❖ in situ stress and induced stress measurements
- ❖ Hydrological test for water flow properties, seepage pressure and permeability tests
- ❖ Dynamic properties of rockmass
- ❖ Deformation modulus of rockmass



Typical stress versus deformation curve recorded in a deformability test of a rock mass

INTRODUCTION

- *Modulus of elasticity or Young's modulus (E):* The ratio of stress to corresponding strain below the proportionality limit of a material.
- *Modulus of deformation of a rock mass (E_m):* The ratio of stress (p) to corresponding strain during loading of a rock mass, including elastic and inelastic behaviour (w_d)
- *Modulus of elasticity of a rock mass (E_{em}):* The ratio of stress (p) to corresponding strain during loading of a rock mass, including only the elastic behaviour (w_e)

INTRODUCTION

- The modulus of deformation of rock mass is calculated by taking total deformation of loading cycle at a particular applied stress level.
- The modulus of elasticity of rock mass is calculated by considering deformation of unloading cycle at a particular applied stress level.

DIRECT METHODS

The modulus of deformation of rock mass is determined by conducting the following in-situ tests:

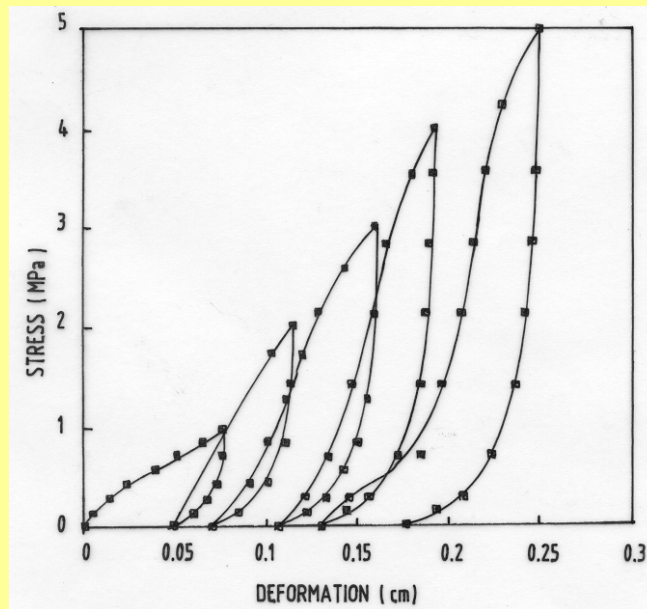
- Plate Jacking Test (PJT) With Borehole Extensometers,
- Plate Loading Test (PLT) With Surface Measurement,
- Flat Jack Test (FJT),
- Radial Jack Test (RJT),
- Dilatometer Test (DT) And
- Goodman Jack Tests (GJT).

Plate loading test assembly at site

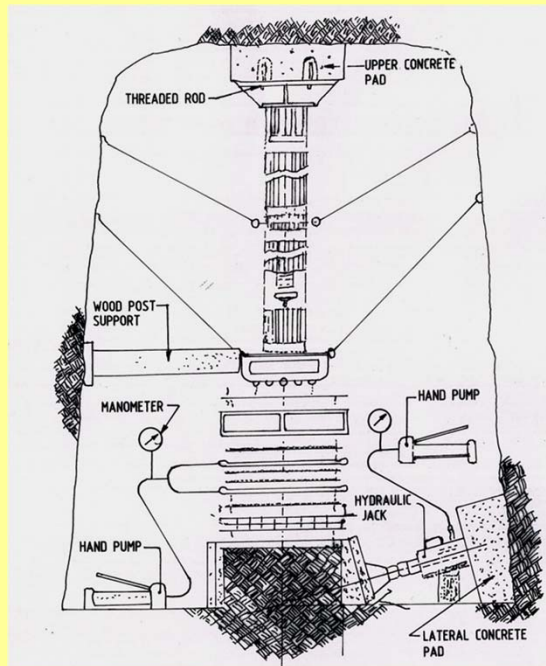


- Plate
- Loading Assembly
- LVDT or Dial Gauges
- Read Out Unit
- Hydraulic Pump

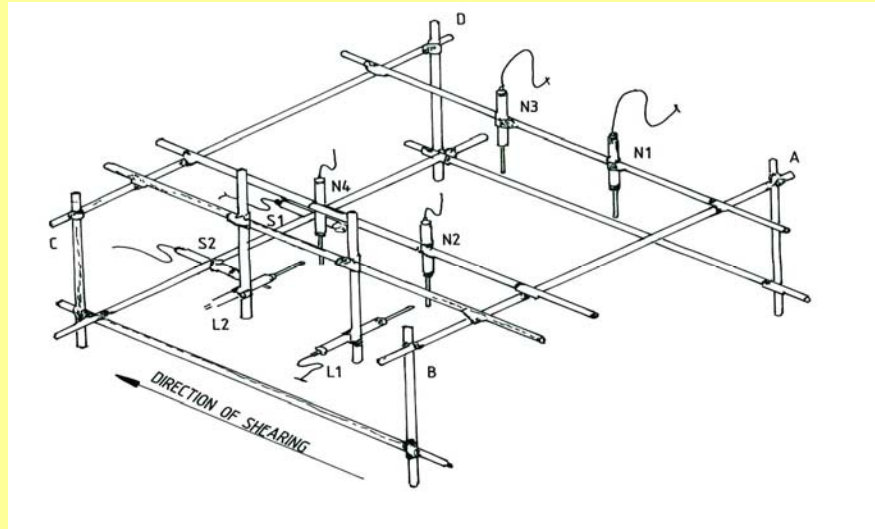
Stress versus Deformation Curve from Plate Loading Test



In-situ shear test inside the drift



Measurements of Normal and Shear Displacements



Shear Strength

- The shear strength between two planes can be measured by the following equation:

$$\tau = c + \sigma \tan \phi$$

where,

- τ = Shear strength between two planes,
- c = Cohesion,
- σ = Normal stress acting on two planes, and
- ϕ = Friction angle.

CALCULATIONS

$$\tau = c + \sigma \tan \phi$$

Normal stress and shear stress are obtained from Normal load and shear load obtained during the test from 5-6 blocks.

The Shear Stress and Normal Stresses are calculated from the following equations:

$$\text{Shear Stress, } \tau = \frac{P_s}{A} = \frac{P_{sa} \cos \alpha}{A}$$

$$\text{Normal stress, } \sigma_n = \frac{P_n}{A} = \frac{P_{na} + P_{sa} \sin \alpha}{A}$$

Where,

Psa = Total applied Shear Load

Ps = Total shear force

Pn = Total normal force

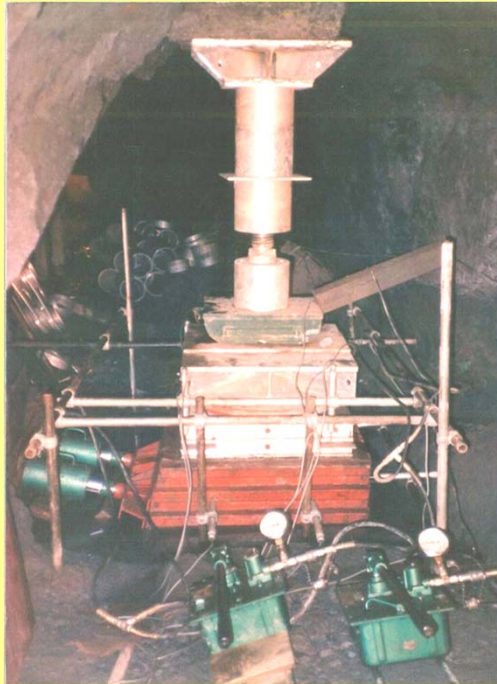
A = Area of test block

Pna = Total applied Normal Load

α = Inclination of Applied shear force

The shear stress is applied at an angle of 15° to the horizontal to avoid turning moment of block during shearing.

**In-situ Shear
Test Assembly**



In-situ Shear Test



Conforms with ISRM “Suggested Methods”